ConnectAI CRM - Complete User Guide & Operating Handbook

Application: ConnectAI CRM - Omnichannel WhatsApp, Social & Voice AI SaaS
Android Package ID: com.connectai.crm.mobile (Debug: com.connectai.crm.mobile.debug)
Architecture: React 19 + Express + Cloud SQL PostgreSQL (europe-west1) + Server-Side Google Gemini AI + Native Android (Kotlin 2.0 & Jetpack Compose Material 3)

Table of Contents

1. [Chapter 1 - What ConnectAI CRM Does & System Overview](#chapter-1--what-connectai-crm-does--system-overview)
2. [Chapter 2 - Installing & Opening the Android App](#chapter-2--installing--opening-the-android-app)
3. [Chapter 3 - First-Time Workspace Setup & Connecting the Live Backend URL](#chapter-3--first-time-workspace-setup--connecting-the-live-backend-url)
4. [Chapter 4 - Understanding the Overview Dashboard](#chapter-4--understanding-the-overview-dashboard)
5. [Chapter 5 - Managing Contacts & Duplicate Consolidation (Lead Center)](#chapter-5--managing-contacts--duplicate-consolidation-lead-center)
6. [Chapter 6 - International Recruitment Pipeline (11 Stages)](#chapter-6--international-recruitment-pipeline-11-stages)
7. [Chapter 7 - Unified Omnichannel Inbox & Grounded AI Draft Replies](#chapter-7--unified-omnichannel-inbox--grounded-ai-draft-replies)
8. [Chapter 8 - AI Recruitment & Sales Copilot](#chapter-8--ai-recruitment--sales-copilot)
9. [Chapter 9 - WhatsApp Business Cloud API, Social Channels & 12-Step Webhook Simulator](#chapter-9--whatsapp-business-cloud-api-social-channels--12-step-webhook-simulator)
10. [Chapter 10 - Campaign Studio (Click-to-WhatsApp & Social Boosts)](#chapter-10--campaign-studio-click-to-whatsapp--social-boosts)
11. [Chapter 11 - Follow-up Tasks & Candidate Interview Scheduling](#chapter-11--follow-up-tasks--candidate-interview-scheduling)
12. [Chapter 12 - Team Performance, Staff Invitations & RBAC Governance](#chapter-12--team-performance-staff-invitations--rbac-governance)
13. [Chapter 13 - External Integrations & Honest Status Indicators](#chapter-13--external-integrations--honest-status-indicators)
14. [Chapter 14 - Settings, Multi-Tenant Isolation, Security & Clean Workspace Reset](#chapter-14--settings-multi-tenant-isolation-security--clean-workspace-reset)
15. [Chapter 15 - Building & Updating the Android APK / AAB on GitHub Actions](#chapter-15--building--updating-the-android-apk--aab-on-github-actions)
16. [Chapter 16 - Complete Troubleshooting Guide (HTML vs. JSON, Offline Mode,

Build
Errors)](#chapter-16--complete-troubleshooting-guide-html-vs-json-offline-mode-build-errors)

Chapter 1 - What ConnectAI CRM Does & System Overview

ConnectAI CRM is a multi-tenant omnichannel customer relationship management and international recruitment platform built for recruitment agencies, B2B sales teams, and service businesses.

Core Capabilities Implemented in This Project

* Clean, Zero-Fake-Data Workspace Initialization: Every new workspace starts with 0 leads, 0 contacts, 0 conversations, 0 tasks, and $0 pipeline value. No sample or fabricated records are ever inserted into your real business database.
* Overview Dashboard: Displays real-time KPI cards calculated strictly from your stored database records (Total Leads, Active Threads, Qualified Leads, Pipeline Value, Follow-ups Due, and Conversion Rate).
* Lead Center & Duplicate Consolidation: Manage verified contacts across WhatsApp, Instagram, Facebook, and TikTok, with automatic duplicate phone/email detection and one-tap record merging.
* 11-Stage International Recruitment Pipeline: Track candidates across New Lead, Contacted, Interested, Documents Pending, Qualified, Employer Review, Offer, Visa Process, Deployed, Rejected, and On Hold.
* Unified Omnichannel Inbox: Read and reply to customer messages, add yellow internal staff notes, and generate grounded AI reply drafts using server-side Google Gemini.
* WhatsApp & Channels + 12-Step AI Qualification Engine: Register official WhatsApp Business Cloud API numbers or test end-to-end automated lead capture and scoring using the built-in 12-Step Inbound Webhook Simulator.
* Follow-up Tasks & Candidate Interviews: Assign due dates, track overdue items, and book candidate or employer interviews.
* Team RBAC & Security Audit Trail: Enforce 5 role levels (company_owner, manager, analyst, agent, viewer), control CSV data export permissions, trigger an Emergency AI Automation Pause, and inspect immutable security audit logs.

Chapter 2 - Installing & Opening the Android App

Step-by-Step Installation on Your Android Phone

1. Open your GitHub repository (alammustak355-afk/Tabita-Ghimire) in your mobile browser and tap Actions.
2. Tap the latest green workflow run titled Build ConnectAI CRM Android APK & Play

Store AAB.
3. Scroll down to the Artifacts section and tap connectai-crm-debug-apk to download the ZIP archive.
4. Open your phone's Files / Downloads app, tap the downloaded ZIP file, and extract app-debug.apk.
5. Tap app-debug.apk to install ConnectAI CRM (com.connectai.crm.mobile.debug). If prompted by Android, allow installation from your browser or file manager.
6. Tap Open to launch ConnectAI CRM.

Chapter 3 - First-Time Workspace Setup & Connecting the Live Backend URL

Why the App Shows "Offline / Backend Not Connected" Until Cloud Run Is Deployed

* Google AI Studio's preview URLs (https://ais-pre-...run.app and https://ais-dev-...run.app) are protected by an interactive browser cookie gate (/__cookie_check.html).
* When a native Android app calls an ais-pre-... preview URL, Google's proxy responds with HTTP 302 or text/html (<!doctype html>) instead of JSON.
* ConnectAI CRM safely intercepts HTML and redirect responses before JSON parsing so the app never crashes, and displays a clear Backend API Diagnostics & Configuration dialog.

How to Deploy & Connect Your Live Backend URL

1. In Google AI Studio, tap the Deploy / Publish button (rocket icon) at the top right to deploy this project to a public Google Cloud Run service.
2. Copy the deployed Cloud Run HTTPS URL (for example: https://connectai-crm-xxxxxx.europe-west2.run.app).
3. Open the ConnectAI CRM Android app on your phone.
4. Tap Set API URL / Diagnostics on the top banner (or open the left drawer -> Settings & Security).
5. Paste your deployed Cloud Run URL into Public Cloud Run HTTPS Backend URL (the app automatically normalizes the URL, forces https://, and strips accidental /api/bootstrap suffixes).
6. Tap Test URL or Save & Retry Live API.
7. Once connected, the top status bar turns emerald green: LIVE CLOUD SQL * <Your Workspace Name>.

Chapter 4 - Understanding the Overview Dashboard

* Executive Hero Banner: Shows your active workspace name, connection status, and quick buttons for + Add New Lead and Open Unified Inbox.
* Search & Stage Filter Strip: Filter your dashboard metrics in real time by keyword (candidate name, role, destination country) or by recruitment stage.
* 6 Live KPI Cards:
* Total Leads: Count of active (non-archived) leads/candidates and total CRM contacts.
* Active Threads: Open omnichannel conversations and unread message count.
* Qualified Leads: Candidates in Qualified, Employer Review, Offer, Visa Process, Deployed, or Won.
* Pipeline Value: Sum of estimated placement value (USD) for qualified leads.
* Follow-ups Due: Count of pending follow-up tasks and scheduled interviews.
* Conversion Rate: Percentage of active leads that have reached Offer, Visa Process, Deployed, or Won.

Chapter 5 - Managing Contacts & Duplicate Consolidation (Lead Center)

Adding a New Contact

1. Open Lead Center from the bottom bar or navigation drawer.
2. Tap + Contact.
3. Enter Full Name and Phone / WhatsApp Number (required), plus optional Email, Employer/Company, and Requirements Summary.
4. Tap Save Contact to PostgreSQL.

Merging Duplicate Contacts

* Whenever two contacts share the same normalized phone number (7+ digits) or email address, a yellow Detected Duplicate Contacts banner appears automatically at the top of Lead Center.
* Tap Merge Records next to any pair to consolidate their conversations, leads, and notes into the primary contact record and log an audit entry.

Chapter 6 - International Recruitment Pipeline (11 Stages)

Adding a Candidate or Business Lead

1. Tap the + icon in the top bar (or + Add Candidate inside Recruitment Pipeline).
2. Fill in the candidate’s Name, Phone/WhatsApp, Email, Nationality, Preferred Job / Role, Destination Country, Placement Value (USD), and Qualifications & Notes.
3. Tap Save to Workspace.

Moving Candidates Across the 11 Stages

* Each candidate card includes a horizontal Quick Stage Selector Strip:
New Lead -> Contacted -> Interested -> Documents Pending -> Qualified -> Employer Review -> Offer -> Visa Process -> Deployed -> Rejected -> On Hold.
* Tap any stage chip on a candidate card to immediately update their stage in PostgreSQL.
* Tap Archive to hide inactive candidates without losing historical data, or Delete (restricted to authorized RBAC roles) to permanently remove a record.

Chapter 7 - Unified Omnichannel Inbox & Grounded AI Draft Replies

1. Open Unified Inbox from the bottom navigation bar.
2. Tap any conversation chip along the top horizontal bar to load the full message history.
3. Generating an AI Reply Draft:
* Tap AI Draft Reply at the top right of the thread, or tap Use Suggested Reply inside the blue AI Summary box.
* The draft is placed into your message input box so you can review or edit it before sending.
4. Sending a Reply vs. Internal Staff Note:
* Keep the chip set to Reply to send an outbound customer message.
* Tap the chip to switch to Internal Note (highlighted in amber) to record a private staff-only comment inside the thread.

Chapter 8 - AI Recruitment & Sales Copilot

1. Open AI Copilot from the bottom navigation bar or left drawer.
2. Select one of the 4 grounded analysis modes:
* Candidate Qualification: Scores candidate readiness and missing compliance documents.
* Executive Daily Briefing: Summarizes urgent threads, overdue tasks, and pipeline value.
* Draft WhatsApp Follow-Ups: Generates personalized WhatsApp templates for active candidates.
* Pipeline Bottleneck Audit: Identifies stages where candidates are stalling.
3. Optionally customize the instruction prompt and tap Run AI Copilot Analysis.
4. Note: AI Copilot runs server-side via /api/ai/copilot using the server’s GEMINI_API_KEY. It requires your live backend to be connected.

Chapter 9 - WhatsApp Business Cloud API, Social Channels & 12-Step Webhook Simulator

1. Open WhatsApp & Channels from the navigation drawer.
2. Connecting a Business Line: Enter your Display Name, Phone Number / @Handle, and Meta Phone Number ID, then tap Register Channel Endpoint.
3. Testing Inbound Messages Without Meta API Credentials (12-Step AI Engine):
* Scroll to the Test Inbound Candidate Message (12-Step AI Engine) card.
* Enter a candidate name, phone number, and sample inbound message (e.g., *"Hi, I am a qualified TIG Welder looking for sponsorship"*).
* Tap Simulate Inbound Webhook + AI Qualification.
* The server automatically creates/updates the contact, opens a conversation thread, scores the lead, extracts job requirements, drafts a grounded reply, and creates a follow-up task.

Chapter 10 - Campaign Studio (Click-to-WhatsApp & Social Boosts)

1. Open Campaign Studio from the navigation drawer.
2. Choose a target channel (INSTAGRAM, FACEBOOK, TIKTOK, or WHATSAPP).
3. Enter your Campaign Title, Campaign Objective, Ad Copy & Call-to-Action, and Daily Budget (USD).
4. Tap Create Campaign & Queue Approval to save the campaign in your workspace.

Chapter 11 - Follow-up Tasks & Candidate Interview Scheduling

1. Open Follow-up Tasks from the navigation drawer.
2. Create a Task: Enter the Task Description, Candidate/Contact name, and Due Date (YYYY-MM-DD), then tap Save Follow-up Task.
3. Complete a Task: Tap the checkbox next to any task to toggle its status between pending and completed.
4. Book an Interview: Scroll to + Book Candidate / Employer Interview, enter the Candidate/Employer Name, Interview Title, and Date & Time (UTC), and tap Schedule Interview.

Chapter 12 - Team Performance, Staff Invitations & RBAC Governance

1. Open Team & Performance from the navigation drawer.
2. Staff Scorecard: View each team member’s active lead count and placed/won candidate count.
3. Active Session RBAC Role Switcher: Test or enforce role permissions across company_owner, manager, analyst, agent, and viewer, and toggle Grant CSV Data Export Permission.
4. Invite Team Member: Enter a staff email address and department, then tap Send Workspace Invitation.

Chapter 13 - External Integrations & Honest Status Indicators

Open Integrations from the navigation drawer to inspect the real status of external providers:
* Meta WhatsApp Business Cloud API: Shows CONNECTED only when META_WHATSAPP_ACCESS_TOKEN is configured on the server; otherwise operates in verified Webhook Simulator mode.
* Google Gemini AI: Processes grounded AI Copilot and reply drafting server-side so no API key is ever exposed inside the Android APK.

Chapter 14 - Settings, Multi-Tenant Isolation, Security & Clean Workspace Reset

Open Settings & Security from the navigation drawer to:
* Switch Tenant Workspaces: Switch between isolated company workspaces (tenant_id).
* Emergency AI Automation Pause: Flip the master kill-switch to immediately halt all automated AI responses and route all inbound messages to human staff.
* Configure Production Backend API Endpoint: Test and save your public Cloud Run HTTPS URL.
* Inspect Immutable Security Audit Logs: Review timestamped records of logins, RBAC changes, contact merges, and configuration updates.

Chapter 15 - Building & Updating the Android APK / AAB on GitHub Actions

Whenever code changes are pushed to the main branch of your GitHub repository

(alammustak355-afk/Tabita-Ghimire), the workflow
.github/workflows/android-release.yml automatically:
1. Sets up JDK 17 (Temurin) and Android SDK Platform 35 (compileSdk = 35, targetSdk = 35, minSdk = 26).
2. Runs ./gradlew assembleDebug --stacktrace to produce the installable phone APK (connectai-crm-debug-apk).
3. Runs ./gradlew bundleRelease --stacktrace to produce the Play Store Android App Bundle (connectai-crm-release-aab).

Chapter 16 - Complete Troubleshooting Guide (HTML vs. JSON, Offline Mode, Build Errors)

Problem 1: 'Value <!doctype ... of type java.lang.String cannot be converted to JSONObject'

* Root Cause: Older builds followed a 302 redirect from the AI Studio preview URL (ais-pre-...run.app) to /__cookie_check.html and passed the HTML login page directly to JSONObject(...).
* How It Is Fixed: ConnectAiApiClient.kt now disables automatic redirects (instanceFollowRedirects = false), inspects HTTP status codes (300..399) and Content-Type headers before parsing, and never crashes on HTML responses. To switch from Offline Mode to Live Mode, deploy the backend to a public Cloud Run URL and paste that URL in Set API URL / Diagnostics.

Problem 2: GitHub Actions Run #3 Failed with Exit Code 1 ('build-android')

* Root Cause: In commit def213ea415c4c11bf99f4756c3a03b4bc5ecd40, NavDestination.HELP_USER_GUIDE was added to enum class NavDestination in CrmModels.kt, but MainActivity.kt did not include a NavDestination.HELP_USER_GUIDE -> ... branch in its when (activeDest) block. Kotlin 2.0 requires when expressions over enums to be exhaustive.
* How It Is Fixed: MainActivity.kt now includes the NavDestination.HELP_USER_GUIDE -> HelpUserGuideScreen(...) branch, and OperationsAndAdminScreens.kt implements HelpUserGuideScreen so all 13 navigation destinations compile cleanly.

Problem 3: Workspace Shows Zero Leads / Zero Pipeline Value After Connecting

* Explanation: By design, ConnectAI CRM starts 100% clean with zero fake sample records. Tap the + button in the top bar (or use the 12-Step Inbound Webhook Simulator in WhatsApp & Channels) to create your first real candidate or lead.